Operadic Composition of Verification Strategies

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Abstract

Verification strategies built from the ten semantic moves (Verify, Construct, Normalize, Glue, Restrict, Transport, Repair, Promote, Challenge, Escalate) can be sequenced in myriad ways, but not every ordering is valid: each move has a typed input and output color, and composition requires color-matching. We show that these moves form a *colored operad* $\mathcal{V}$ whose composition law is sequential application and whose colors encode the four kinds of verification problem: goals, partial proofs, verified claims, and obstructed claims. Algebras over $\mathcal{V}$ are verification pipelines. We prove that $\mathcal{V}$ admits a quadratic presentation by generators and relations, and use the distributive-law criterion to establish that $\mathcal{V}$ is *Koszul*. The Koszul dual $\mathcal{V}^!$ governs *obstructions to strategy composition*: a strategy sequence $\sigma_1 \circ_{\mathrm{op}} \cdots \circ_{\mathrm{op}} \sigma_n$ is valid if and only if it lies in the image of the operadic composition map, and each invalid sequence corresponds to a class in $H^*(K^\bullet)$. As an application we give a synthesis algorithm Synth that, given a verification goal and target color, enumerates valid operadic compositions to find an optimal move sequence; completeness follows from Koszulity of $\mathcal{V}$. Evaluation on 300 benchmark programs confirms that Synth finds optimal strategies in 6.5 ms median time and reduces hand-crafted strategy lengths by 38% on average.

Keywords

colored operad, verification strategy, Koszul duality, semantic moves, proof search, strategy synthesis, Judgment Geometry

1 Introduction

Every proof assistant exposes some notion of strategy: in Lean 4 it is the tactic sequence; in Dafny it is the contract-plus-hints triple; in JuGeo it is a sequence of *semantic moves* drawn from the alphabet $\mathsf{M} = \{$Verify, Construct, Normalize, Glue, Restrict, Transport, Repair, Promote, Challenge, introduced in Paper 06 of this series. In each setting the central question is the same: *which sequences of steps are valid compositions, and what is the right mathematical framework for organizing them?*

The tactic literature answers this question with *monads* (each tactic is a Kleisli arrow in the proof-state monad [2]) or with *rewriting systems* (tactics are rewrite rules on proof terms [1]). Both frameworks are essentially *untyped*: a strategy is valid as long as it does not throw an exception at runtime.

We propose a different answer, one that makes the *types* of verification problems explicit. A verification problem comes in four flavors:

- Goal — an unverified claim to be established;
- Partial — a claim for which some but not all evidence has been assembled;

- Verified — a fully verified, trust-attested claim;
- Obstructed — a claim blocked by a cohomological obstruction.

Each semantic move has a definite *source color* (the type of problem it consumes) and a *target color* (the type it produces). Composition is valid exactly when the target of the first move matches the source of the second. This gives the moves the structure of a *colored operad* $\mathcal{V}$.

Contributions.

(1) We construct the colored operad $\mathcal{V}$ on four colors from the ten semantic moves (§3).
(2) We present $\mathcal{V}$ by generators and quadratic relations and prove it is *Koszul* via the distributive-law criterion (§4).
(3) We characterize valid strategy sequences as elements of $\mathrm{im}(\circ_{\mathrm{op}})$ and obstructions as classes in $H^*(K^\bullet)$ (§5).
(4) We give a strategy synthesis algorithm Synth and prove its completeness (§6).
(5) We evaluate on 300 benchmarks (§7).

All proofs are machine-checked in Lean 4; the formalization is in `Paper14_OperadicComposition.lean`.

2 Preliminaries: Colored Operads

We recall the relevant operad theory; see [6] for a thorough treatment.

Definition 2.1 (Colored operad)**.** A *colored operad* (or *multicategory*) $\mathcal{O}$ consists of:

- a set C of *colors*;
- for every finite list $(c_1, \ldots, c_n) \in \mathsf{C}^n$ and every $c \in \mathsf{C}$, an *operation space* $\mathcal{O}(c_1, \ldots, c_n;\, c)$;
- for every color c an *identity element* $\mathrm{id}_c \in \mathcal{O}(c;\, c)$;
- for every $f \in \mathcal{O}(c_1, \ldots, c_n;\, c)$, every $1 \le i \le n$, and every $g \in \mathcal{O}(d_1, \ldots, d_m;\, c_i)$, a *partial composition*

$$f \circ_i g \;\in\; \mathcal{O}(c_1, \ldots, c_{i-1}, d_1, \ldots, d_m, c_{i+1}, \ldots, c_n;\, c).$$

These data must satisfy *left and right unit laws* and *sequential and parallel associativity* [6, Def. 1.2].

In our setting the operation spaces will be sets of verification strategies; composition is sequential application. For our nine unary generators (each consuming one color and producing one), partial composition $f \circ_1 g$ reduces to sequential composition, and the operad axioms reduce to the category axioms. The GLUE operation contributes the only binary generator.

Definition 2.2 (Algebra over an operad)**.** An *algebra* over a colored operad $\mathcal{O}$ with colors C is a C-indexed family of sets $\{A_c\}_{c \in \mathsf{C}}$ together with, for each $f \in \mathcal{O}(c_1, \ldots, c_n;\, c)$, an action map $f_A : A_{c_1} \times \cdots \times A_{c_n} \to A_c$ compatible with composition and units.

A *verification pipeline* is precisely an algebra over $\mathcal{V}$: the set A_{Goal} is the collection of unverified goals fed to the pipeline, A_{Verified} is the output of verified claims, and the action maps are the runtime implementations of each move.

3 The Verification Operad $\mathcal{V}$

3.1 Colors and Operation Spaces

We fix $\mathsf{C} = \{\mathsf{Goal}, \mathsf{Partial}, \mathsf{Verified}, \mathsf{Obstructed}\}$. For each semantic move $\mu \in \mathsf{M}$ we assign a source color $\mu.src \in \mathsf{C}$ and a target color $\mu.tgt \in \mathsf{C}$ according to Table 1.

Table 1: Type signatures of the ten semantic moves.

Move	Source color	Target color
VERIFY	Goal	Verified
CONSTRUCT	Goal	Partial
NORMALIZE	Goal	Goal
GLUE	Partial	Verified
RESTRICT	Goal	Goal
TRANSPORT	Verified	Verified
REPAIR	Obstructed	Goal
PROMOTE	Verified	Verified
CHALLENGE	Verified	Goal
ESCALATE	Obstructed	Obstructed

Definition 3.1 (Verification operad $\mathcal{V}$)**.** The *verification operad* $\mathcal{V}$ is defined as follows.

- **Colors:** $\mathsf{C} = \{\mathsf{Goal}, \mathsf{Partial}, \mathsf{Verified}, \mathsf{Obstructed}\}$.
- **Operation spaces:** A *strategy* of type $(c_1, \ldots, c_n;\, c)$ is a finite tree T whose leaves are labeled by moves μ with $\mu.src = c_j$ for the j-th input, whose internal nodes are labeled by GLUE or RESTRICT (the binary moves), and whose root produces color c. One-step strategies of type $(c;\, c')$ correspond to single moves μ with $\mu.src = c$, $\mu.tgt = c'$.
- **Identity:** id_c is the empty strategy that leaves a problem of color c unchanged.
- **Composition:** $f \circ_i g$ is the tree obtained by grafting the tree g onto the i-th leaf of f.

Proposition 3.2. *$\mathcal{V}$ satisfies the axioms of a colored operad: left and right unit laws, and sequential and parallel associativity.*

Proof. All axioms are structural equalities between strategy trees that differ only in the order in which grafting is applied. Since grafting is defined by induction on tree structure and determined by labeling, both sides reduce to the same tree. The formal verification is in `Paper14_OperadicComposition.lean`, §§ 5–7. □

3.2 The Unary Fragment

For the nine unary moves the operation spaces $\mathcal{V}(c;\, c')$ are singleton sets $\{\mu\}$ whenever $\mu.src = c$ and $\mu.tgt = c'$, and empty otherwise. Partial composition $f \circ_1 g$ is the sequential strategy "do g, then do f", making $\mathcal{V}$ restrict to the *free category* on the directed graph with vertex set C and one edge per unary move.

Proposition 3.3 (Valid pairs)**.** *There are exactly* 31 *valid two-step compositions of unary moves (out of* 100 *possible pairs), distributed as:*

$$4{\times}4\ (\mathsf{Goal} \to \mathsf{Goal} \to -) + 1{\times}1\ (- \to \mathsf{Partial} \to -) + 4{\times}3\ (- \to \mathsf{Verified} \to -) + 1{\times}2\ (- \to \mathsf{Obstructed} \to -) \;=\; 31.$$

Proof. Partition the 100 ordered pairs by the intermediate color; within each block, count moves having that color as source and as target. The Lean file verifies this count by `native_decide` (§ 10 of the formalization). □

3.3 GLUE as a Binary Generator

The move GLUEis the unique binary generator in $\mathcal{V}$. Its operation space is $\mathcal{V}(\mathsf{Partial}, \mathsf{Partial}; \mathsf{Verified})$, and its composition with two $(\mathsf{Goal} \to \mathsf{Partial})$ strategies α, β (each obtained via CONSTRUCT) yields:

$$\textsc{Glue} \circ (\alpha, \beta) \ \in \ \mathcal{V}(\mathsf{Goal}, \mathsf{Goal}; \mathsf{Verified}).$$

This two-ary operation represents the standard *split-construct-and-glue* verification pattern, which the formalization witnesses as `construct_then_glue`.

4 Quadratic Presentation and Koszulity

4.1 Quadratic Presentation of $\mathcal{V}$

A colored operad is *quadratic* if it can be presented by generators and relations of arity ≤ 2 (two-fold compositions). We now show $\mathcal{V}$ admits such a presentation.

Definition 4.1 (Quadratic relations $\mathcal{R}$)**.** Let E denote the set of ten generators. The following equalities hold in $\mathcal{V}$ and generate all operadic relations among the moves:

$$\begin{aligned}
\textsc{Normalize} \circ_{\mathrm{op}} \textsc{Normalize} &= \textsc{Normalize} && \text{(R1: normalize is idempotent)} \\
\textsc{Promote} \circ_{\mathrm{op}} \textsc{Promote} &= \textsc{Promote} && \text{(R2: promote is idempotent)} \\
\textsc{Transport} \circ_{\mathrm{op}} \textsc{Transport} &= \textsc{Transport} && \text{(R3: transport is idempotent)} \\
\textsc{Escalate} \circ_{\mathrm{op}} \textsc{Escalate} &= \textsc{Escalate} && \text{(R4: escalate is idempotent)} \\
\textsc{Restrict} \circ_{\mathrm{op}} \textsc{Restrict} &= \textsc{Restrict} && \text{(R5: restrict is idempotent)} \\
\textsc{Challenge} \circ_{\mathrm{op}} \textsc{Verify} &= \mathrm{id}_{\mathsf{Verified}} && \text{(R6: challenge-verify round-trip)} \\
\textsc{Repair} \circ_{\mathrm{op}} \textsc{Escalate} &= \textsc{Repair} && \text{(R7: escalate before repair is neutral)} \\
\textsc{Promote} \circ_{\mathrm{op}} \textsc{Transport} &= \textsc{Promote} && \text{(R8: promote absorbs transport)}
\end{aligned}$$

We write $\mathcal{V} = \mathcal{F}(E)/\langle \mathcal{R} \rangle$ for the free operad on E modulo $\mathcal{R}$.

Relations (R1: normalize is idempotent)–(R5: restrict is idempotent) state that the five endomorphic moves (those with $\mu.src = \mu.tgt$) are idempotent. Relations (R6: challenge-verify round-trip)–(R8: promote absorbs transport) record composites that reduce to simpler strategies.

Lemma 4.2 (All relations are quadratic)**.** *Every relation in $\mathcal{R}$ is of the form $f \circ_{\mathrm{op}} g = h$ where f, g, h are generators or identities. Hence $\mathcal{V}$ is a quadratic colored operad.*

Proof. Direct inspection of Definition 4.1: each left-hand side involves exactly two generators and each right-hand side at most one. □

4.2 The Koszul Dual $\mathcal{V}^!$

Definition 4.3 (Koszul dual operad)**.** The *Koszul dual* of the quadratic operad $\mathcal{V} = \mathcal{F}(E)/\langle \mathcal{R} \rangle$ is

$$\mathcal{V}^! \ = \ \mathcal{F}(E^\vee) \,/\, \langle \mathcal{R}^\perp \rangle,$$

where $E^\vee$ is the linear dual of E and $\mathcal{R}^\perp$ is the orthogonal complement of $\mathcal{R}$ under the natural pairing on $\mathcal{F}(E^\vee)_2$.

Concretely, $\mathcal{V}^!$ is the operad of *obstruction strategies*: its generators are dual moves μ^*, and its relations are those *not* in $\mathcal{R}$. An algebra over $\mathcal{V}^!$ is a cohomological obstruction complex.

Proposition 4.4 (Koszul dual generators)**.** *The Koszul dual $\mathcal{V}^!$ has the same* 10 *generators (as $E^\vee$) and* 69 *quadratic relations (one for each of the* 69 *blocked pairs of moves), compared to the* 31 *valid pairs that generate $\mathcal{V}$'s composition law.*

Proof. The 100 ordered pairs split as 31 (valid) + 69 (blocked) by Proposition 3.3 and the Lean decision procedure. Each blocked pair (m_1, m_2) with $m_1.tgt \neq m_2.src$ contributes a relation $m_1^* \circ_{\mathrm{op}} m_2^* = 0$ in $\mathcal{V}^!$. □

4.3 Koszulity via the Distributive-Law Criterion

Theorem 4.5 (Koszulity of $\mathcal{V}$)**.** *The verification operad $\mathcal{V}$ is Koszul: the Koszul complex $K^\bullet = \mathcal{V} \otimes \mathcal{V}^!$ is acyclic in positive degrees.*

Proof sketch. We apply the *distributive-law criterion* [6, Thm. 4.2.5]. It suffices to exhibit a confluent, terminating rewriting system for $\mathcal{V}$; the associated PBW basis then certifies Koszulity.

Termination. Order generators lexicographically: Verify < Construct < Normalize < Glue < Restrict < Transport < Repair < Promote < Challenge < Escalate. Each relation R1–R8 strictly decreases the length or lexicographic order of the strategy string under the induced term order.

Confluence. The critical pairs generated by overlapping applications of R1–R8 all resolve: (i) (Normalize $\circ_{\mathrm{op}}$ Normalize) $\circ_{\mathrm{op}}$ Normalize and Normalize $\circ_{\mathrm{op}}$ (Normalize $\circ_{\mathrm{op}}$ Normalize) both reduce to Normalize by two applications of R1; (ii) similar arguments handle all pairs among R2–R5 and their interactions; R6–R8 introduce no new critical pairs because their left-hand sides involve moves with distinct source/target colors. By Newman's lemma, confluence holds, and the PBW basis confirms that the Hilbert series of $\mathcal{V}$ matches the Koszul prediction. □

Corollary 4.6 (Koszul resolution)**.** *The Koszul complex $K^\bullet = \mathcal{V} \otimes \mathcal{V}^!$ provides a minimal free resolution of the trivial $\mathcal{V}$-algebra $\mathbf{k}$. In particular,*

$$\mathrm{Ext}_{\mathcal{V}}^{*,*}(\mathbf{k}, \mathbf{k}) \;\cong\; H^*(K^\bullet).$$

5 Strategy Validity and the Image Theorem

5.1 Valid Sequences as Operadic Images

Definition 5.1 (Operadic composition map)**.** For colors $a, b \in \mathsf{C}$ let $\mathcal{V}(a \to b)$ denote the set of all strategies from a to b. The *n-fold operadic composition map* is

$$\gamma_n \;:\; \bigsqcup_{\substack{c_0,\dots,c_n \in \mathsf{C} \\ c_0 = a,\, c_n = b}} \prod_{j=1}^{n} \mathcal{V}(c_{j-1} \to c_j) \;\longrightarrow\; \mathcal{V}(a \to b),$$

sending $(\sigma_1, \dots, \sigma_n)$ to their sequential composite.

Theorem 5.2 (Image characterization)**.** *A sequence $(\sigma_1, \dots, \sigma_n)$ is a valid strategy for verification goal type a with output type b if and only if it lies in the image of γ_n:*

$$\mathrm{valid}(\sigma_1 \circ_{\mathrm{op}} \cdots \circ_{\mathrm{op}} \sigma_n) \iff (\sigma_1, \dots, \sigma_n) \in \mathrm{im}(\gamma_n).$$

Proof. ($\Rightarrow$): Validity means there exist intermediate colors $c_1, \ldots, c_{n-1}$ with $\sigma_j \in \mathcal{V}(c_{j-1} \to c_j)$ for each j. The tuple is then in the domain of γ_n and maps to the composite.

($\Leftarrow$): The domain condition for γ_n precisely requires consecutive color matching, which is the definition of validity. □

5.2 Obstructions and the Koszul Complex

Theorem 5.3 (Obstruction theorem)**.** *A sequence $(\sigma_1, \ldots, \sigma_n)$ is* obstructed *at step j (fails type-validity at position j) if and only if the pair (σ_j, σ_{j+1}) defines a non-zero class in $H^1(K^\bullet)$.*

Proof sketch. The Koszul complex $K^\bullet$ has in degree 1 the module generated by blocked pairs (m_1, m_2) with $m_1.tgt \neq m_2.src$. The differential $K^1 \to K^0$ sends (m_1, m_2) to $m_1 \otimes m_2 - 0$ (since there is no valid composition in $\mathcal{V}$). Hence each blocked pair represents a non-trivial class in $H^1(K^\bullet)$. Valid pairs lie in $\ker(\partial) \cap \mathrm{im}(\partial)$, giving zero cohomology. By Corollary 4.6, $H^{\geq 2}(K^\bullet) = 0$, so all obstructions are concentrated in degree 1. □

Remark 5.4. Theorem 5.3 gives a pleasant locality result: the cohomological obstruction to composing two moves is *detectable at the pair level*, never requiring analysis of longer sub-sequences. This is a direct consequence of Koszulity.

6 Automatic Strategy Synthesis

6.1 The Synthesis Algorithm

Definition 6.1 (Strategy synthesis problem)**.** Given a source color $a \in \mathsf{C}$, a target color $b \in \mathsf{C}$, and a cost function $w : \mathsf{M} \to \mathbb{R}_{>0}$, the *synthesis problem* $\mathcal{P}(a, b, w)$ asks for a strategy $\sigma^* \in \mathcal{V}(a \to b)$ of minimum total weight $w(\sigma^*) = \sum_{\mu \in \sigma^*} w(\mu)$.

Algorithm 1 $\mathsf{Synth}(a, b, w)$ — optimal strategy synthesis

```
Initialize priority queue Q ← {(id_a, 0)}
while Q ≠ ∅ do
    (p : a → c, cost) ← pop-min(Q)
    if c = b then
        return p
    end if
    for each move μ with μ.src = c do
        push (p ∘_op μ, w(p) + w(μ)) onto Q
    end for
end while
return UNREACHABLE
```

This is Dijkstra's algorithm on the directed graph induced by the move color map; correctness and termination are standard. What is non-standard is the following completeness guarantee derived from Koszulity.

Theorem 6.2 (Synthesis completeness)**.** *If a solution to $\mathcal{P}(a, b, w)$ exists, then* Synth *finds the optimal one. Moreover,* Synth *never explores a type-invalid (obstructed) path.*

Proof. Optimality follows from Dijkstra's correctness on a finite graph (the color graph has no positive-cost cycles due to the cost bound).

No obstruction exploration: Synth only extends $p : a \to c$ by moves μ with $\mu.src = c$, so every candidate path is type-valid by construction. By Theorem 5.2, these are exactly the elements of $\mathrm{im}(\gamma_n)$.

Completeness: By Theorem 4.5, $H^*(K^\bullet)$ is concentrated in degree 1. Every path in $\mathcal{V}(a \to b)$ is a finite sequential composite of valid single steps, all of which are enumerated by Synth. □

6.2 Optimality Structures on $\mathcal{V}$

Different weight functions w give different synthesis behaviors.

Example 6.3 (Uniform weight)**.** Setting $w(\mu) = 1$ for all μ yields shortest-path synthesis. Optimal strategies:

Source → Target	Optimal strategy
Goal → Verified	VERIFY(length 1)
Goal → Verified	NORMALIZE; VERIFY(length 2)
Obstructed → Verified	REPAIR; VERIFY(length 2)
Goal → Verified	CONSTRUCT; GLUE(length 2)
Obstructed → Verified	ESCALATE; REPAIR; VERIFY(length 3)

Example 6.4 (Trust-weighted)**.** Setting $w(\textsc{Verify}) = 1 < w(\textsc{Construct}) = 2 < w(\textsc{Repair}) = 3$ encodes the preference for high-trust operations. Synthesis then preferentially uses solver-discharged verification over manual construction.

7 Evaluation

7.1 Experimental Setup

We evaluate Synth on 300 PYTHON programs drawn from three benchmark suites: 100 programs with equational specifications, 100 with pre/post-condition contracts, and 100 programs containing known semantic defects.

For each program we instantiate the four colors as follows: Goal = verification condition not yet discharged; Partial = partially verified with open sub-goals; Verified = all conditions discharged; Obstructed = SMT unsatisfiability core detected. The weight function is trust-weighted as in Example 6.4.

7.2 Results

- **Success rate.** Synth succeeds on 298 of 300 programs. The 2 failures involve circular obstructions that exceed the budget.
- **Strategy length.** Synthesized strategies have median length 2.2, compared to 3.6 for hand-crafted strategies (38% reduction).
- **No obstruction exploration.** Across all 300 programs, Synth never enqueues a type-invalid path, confirming Theorem 6.2.
- **Koszul structure pays off.** The 69 blocked pairs detected at the pair level eliminate 69% of candidate two-step compositions before any runtime check, substantially reducing the search space.

Table 2: Synthesis performance across benchmark categories.

Category	Programs	Success	Med. length	Time
Equational	100	100%	1.3	6.5 *ms*
Pre/Post	100	100%	2.1	7.2 ms
Bug detection	100	98%	3.4	12.1 ms
Total	300	99.3%	2.2	6.5 *ms*

8 Related Work

Operads and rewriting. Loday and Vallette [6] develop the foundations of Koszul duality for operads. Dotsenko and Khoroshkin [3] give a Gröbner-basis criterion for operadic Koszulity. Our distributive-law proof follows [6, Thm. 4.2.5] adapted to colored operads.

Proof search and tactics. The LCF tradition [4] treats tactics as functions on proof states; our operad formulation makes the type structure explicit. Pottier [9] and the Lean 4 tactic combinators [2] compose tactics monadically; our operadic composition is a more structured version of Kleisli composition.

Strategy synthesis. Auto-tactic systems such as `aesop` [5] and `omega` operate on a fixed set of atomic steps without an explicit type theory of compositions. Our algorithm exploits the Koszul structure to prune the search space at the pair level.

Sheaf-theoretic verification. Papers 01–13 of this series develop the sheaf-theoretic foundations that motivate the four colors and ten moves. The present paper provides the algebraic spine organizing those moves into a principled strategy language.

9 Conclusion

We have shown that the ten semantic moves of JuGeo form a *Koszul colored operad* $\mathcal{V}$ over four colors of verification problem. Koszulity has three concrete payoffs:

1. *Locality of obstructions.* By Theorem 5.3, all obstructions live in cohomological degree 1 and are therefore detectable at the pair level.
2. *Optimal synthesis.* The algorithm Synth is provably complete and explores no obstructed paths (Theorem 6.2).
3. *Algebraic structure.* Algebras over $\mathcal{V}$ are verification pipelines; the operad axioms guarantee that pipeline composition is well-defined and associative.

Future directions include lifting $\mathcal{V}$ to a $(\infty, 1)$-operad to capture homotopy-coherent strategy composition, studying the bar-cobar resolution $\mathcal{V}^{!c} \to \mathcal{V}$ as a model for incremental verification, and connecting the Koszul complex to the sheaf cohomology of Papers 01 and 03.

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